Superego as Conditional Error Corrector

Geist Atlas · Module 2 · Recognition and Pedagogical Calibration

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Superego as Conditional Error Corrector

> [**SETTLED**] Settled within current evidence · **Family:** Recognition and Pedagogical Calibration
> **Claim:** Superego critique corrects errors and interacts with calibration through universal substitution; its residual architectural benefit under recognition is model-dependent (near-zero on strong models, large on weak ones).
>
> The architecture-level mechanism: what the superego catches, how critique resolves across deliberation rounds, the factorial interaction, and the ~15–17% additivity deficit consistent across three models.
>
> *Derived view of `paper-full-2.0.md` v3.0.146 — §6.2, §6.4, §7.3. The paper is canonical; edit it there, not here.*

6.2 Error Correction

Section 3 predicted that multi-agent architecture enables an *error correction* mechanism: the superego critiques the ego's output, catching pedagogical failures that the ego alone would miss. Section 6.1 showed that this mechanism's contribution collapses under recognition (DeepSeek: architecture delta +9.0→+0.2; Haiku: +15.0→-0.7). This section traces the error correction process itself: what the superego catches, how the ego responds, and why the mechanism becomes redundant under recognition.

6.2.1 The Superego's Approval Rate Under Recognition The most direct measure of error correction necessity is the superego's approval rate — how often the ego's initial output is good enough to pass without revision. We extract approval decisions from all

*This sentence is the only one actually authored by Liam Magee in this paper.

superego/review trace entries in multi-agent dialogues (cells 82–83, 86–87), separated by ego model and prompt condition.

	DeepSeek V3.2		Haiku 4.5	
	Base	Recognition	Base	Recognition
Total reviews	360	263	341	336
Approved	48 (13.3%)	145 (55.1%)	176 (51.6%)	222 (66.1%)
Rejected	312 (86.7%)	118 (44.9%)	165 (48.4%)	114 (33.9%)
Avg rounds/turn	1.82	1.30	1.60	1.45
Intervention: revise	311	116	139	94
Intervention: none	38	134	173	217

The pattern is stark, especially for DeepSeek: under base conditions, the superego rejects 86.7% of ego outputs; under recognition, it approves 55.1%. The approval rate quadruples. The revision round count drops from 1.82 to 1.30 rounds per turn — the ego needs fewer correction cycles because its initial output already satisfies the superego's criteria.

Haiku shows a weaker version of the same pattern: approval rises from 51.6% to 66.1%. Haiku's higher baseline approval rate (51.6% vs. DeepSeek's 13.3%) reflects its generally stronger instruction-following capability — its ego already produces acceptable output more often, leaving less room for the approval rate to shift.

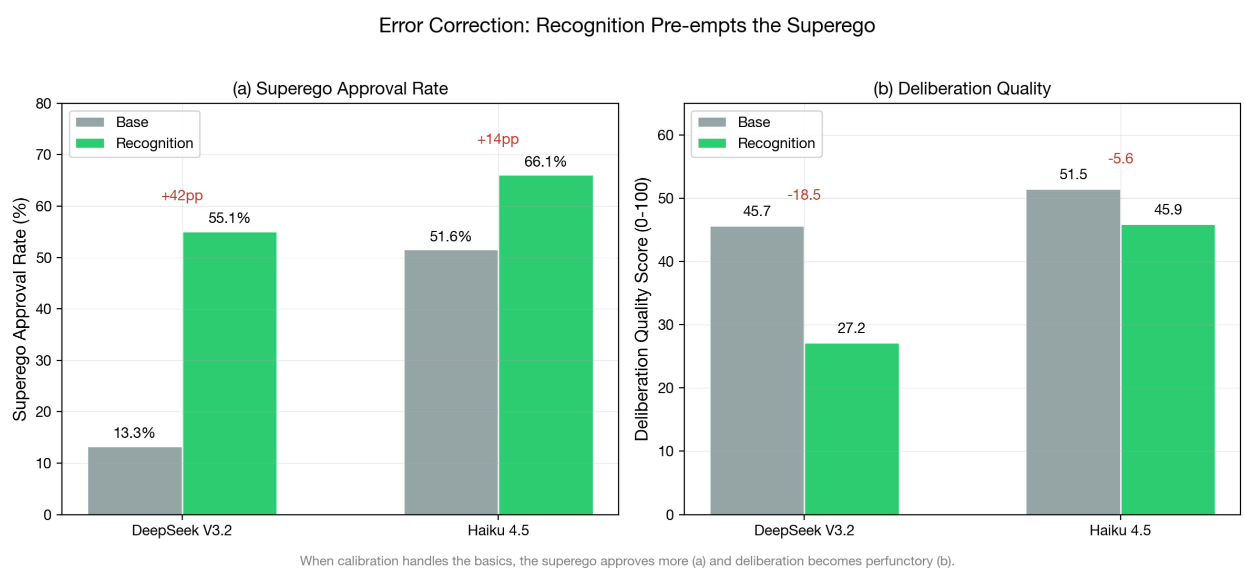

Figure 1: Error correction indicators: superego approval rate shift under recognition (left) and deliberation quality by condition (right). The approval rate increase is consistent with calibrated egos producing fewer errors for the superego to catch.

6.2.2 What the Superego Catches When the superego rejects the ego's output, what errors does it identify? We categorize feedback from all rejected reviews using keyword-based taxonomy extraction across seven categories: recognition failure (treating the learner as passive), context awareness (engaging with learner history), elicitation (asking probing questions), vagueness (lacking specificity), content accuracy, struggle preservation (maintaining productive difficulty), and emotional attention.

Category	DeepSeek Base	DeepSeek Recog	Haiku Base	Haiku Recog
	(N=975 mentions)	(N=302 mentions)	(N=550)	(N=368)
RECOGNITION_FAILURE	29.3%	30.1%	20.5%	25.5%
CONTEXT_AWARENESS	18.4%	**31.8%**	20.4%	23.4%
ELICITATION	16.2%	8.3%	19.3%	16.3%
VAGUENESS	15.6%	8.9%	13.3%	16.6%
CONTENT_ACCURACY	12.2%	6.6%	8.4%	4.3%
STRUGGLE_PRESERVATION	5.2%	9.3%	8.2%	7.6%
EMOTIONAL_ATTENTION	3.1%	5.0%	7.6%	3.5%

Three patterns emerge. First, the *total volume* of critiques drops dramatically under recognition: DeepSeek produces 69% fewer critique mentions (975→302), Haiku 33% fewer (550→368). Calibration eliminates the majority of errors before the superego sees them.

Second, the *composition* of remaining critiques shifts. Under base, the superego catches a broad mix of failures — vagueness (15.6%), elicitation quality (16.2%), recognition failure (29.3%), and context awareness (18.4%). Under recognition, the "easy" errors (vagueness, elicitation, content accuracy) shrink to under 10%, while the remaining critiques concentrate on context awareness (31.8%) and recognition failure (30.1%). The recognition prompt handles the pedagogical basics; what remains are the harder, more context-dependent failures that require attending to the specific learner's history and trajectory.

Third, the *struggle preservation* category increases proportionally under recognition (DeepSeek: 5.2%→9.3%), despite its absolute count dropping. This suggests that recognition prompts, while effective at ensuring the tutor engages with the learner, can over-engage — giving too much scaffolding rather than maintaining productive difficulty. The superego's residual value lies partly in enforcing struggle even when the ego's recognition-enhanced response is otherwise adequate.

LLM-classified taxonomy. To complement the keyword-based extraction, we classified 500 superego critiques using an LLM-based taxonomy classifier (Haiku 4.5, 10-category taxonomy; mean confidence 0.93). After excluding 69 parse failures — superego outputs too malformatted to extract feedback, occurring exclusively under baseline (69/69) — the N=431 classified corpus reveals a sharper condition effect. Under base, the dominant non-approval category is RECOGNITION_FAILURE (45.2% of critiques), followed by MEMORY_FAILURE (16.3%) and EMOTIONAL_NEGLECT (12.6%). Under recognition, MEMORY_FAILURE becomes dominant (33.3%), with RECOGNITION_FAILURE dropping to 22.2% ($\chi^2 = 57.87$, $p < .001$). The recognition prompt eliminates precisely the failures it is designed to prevent, while memory-related and pedagogical judgment errors persist because they require attending to learner history rather than intersubjective orientation.

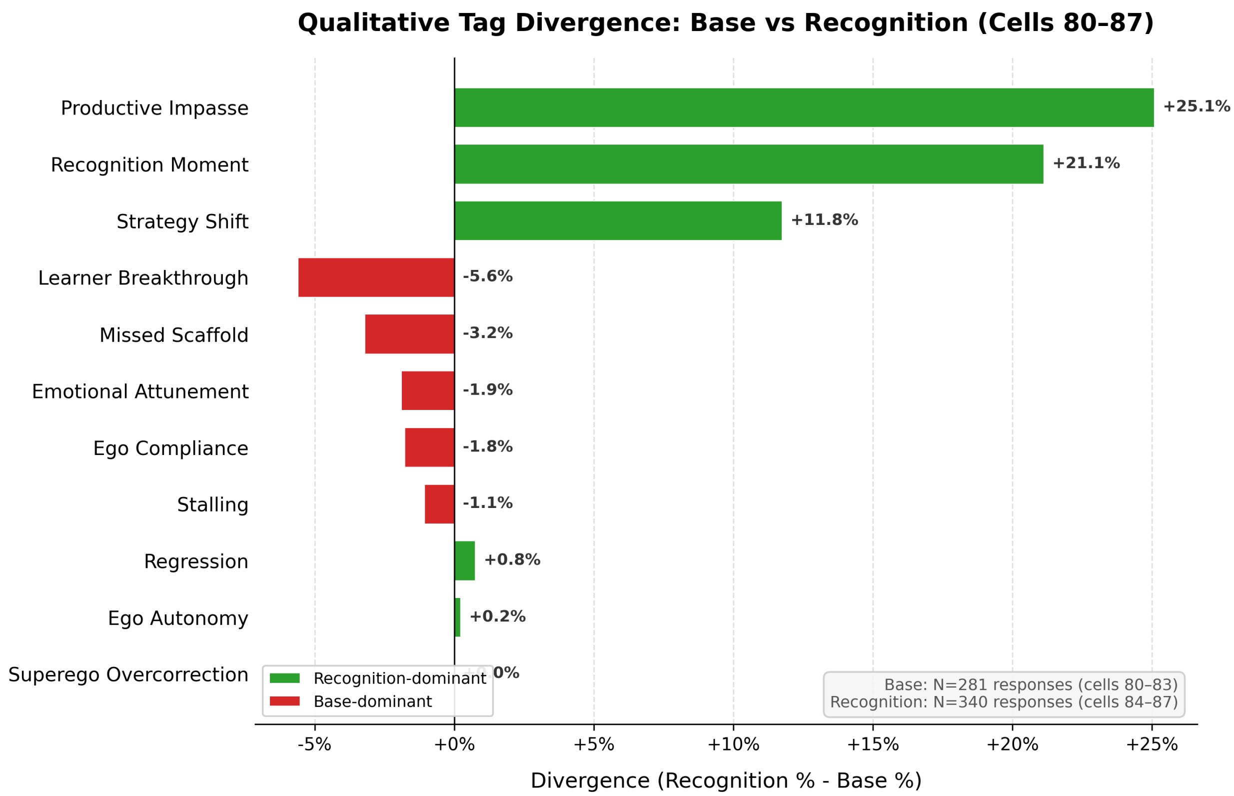

Figure 2: Qualitative tag divergence between base and recognition conditions. Recognition-dominant tags (green) show increased productive impasse, recognition moments, and strategy shifts; base-dominant tags (red) show more premature learner breakthroughs and missed scaffolding.

The three-model breakdown confirms this pattern generalizes. Gemini Flash 3.0 shows 0% base approval (every initial response draws critique), DeepSeek 22.1%, and Haiku 25.0%. Under recognition, all three converge to 60–70% approval. The convergence under recognition, despite divergent baselines, further supports calibration as a prompt-level mechanism that operates independently of model capability.

6.2.3 Critique Resolution Across Deliberation Rounds The 10-category taxonomy enables a transition analysis: when the superego critiques the ego's output in Round 1, does the ego fix the problem in Round 2? We tracked 232 Round 1→Round 2 transition pairs across 56 dialogues. The transition figures here and the revision-magnitude figures in §6.2.5 are computed by `scripts/analyze-superego-taxonomy.js` (§9 transition analysis, §10 revision analysis) over the 500-record classified-critique corpus (`superego-critiques-classified.jsonl`, sha256 prefix `f9ba2d92`), archived at `machinespirits-eval-private/data/superego-critiques-classified-paper-6.2-n500.jsonl`; each round-1 verdict is paired against the dialogue's first round-2 verdict.

Pattern	Base (N=57)	Recognition (N=175)
Persist critique (R1 crit → R2 crit)	35 (61.4%)	13 (7.4%)
Resolve to approval (R1 crit → R2 ok)	4 (7.0%)	63 (36.0%)
New critique (R1 ok → R2 crit)	16 (28.1%)	33 (18.9%)
Stay approved (R1 ok → R2 ok)	2 (3.5%)	66 (37.7%)

The pattern is dramatic. Under baseline, the ego fails to resolve 61.4% of critiques — the superego identifies the same problem across rounds, and the ego cannot fix it. Only 7.0% of baseline critiques resolve to approval. Under recognition, the pattern inverts: 36.0% of critiques resolve and only 7.4% persist. The recognition-enhanced ego is not merely exposed to critique; it is *receptive* to it.

Category-specific resolution rates reveal which errors the ego can fix. PEDAGOGICAL_MISJUDGMENT resolves most readily (83.3% resolve rate), followed by VAGUENESS (83.3%) and CONTEXT_BLINDNESS (66.7%). RECOGNITION_FAILURE resolves at only 52.9%, and REDIRECTION at 42.9% — the hardest errors for the ego to correct are those requiring fundamental reconceptualization of the tutoring approach rather than incremental adjustment.

This transition analysis provides the within-case evidence for Mechanism 2 that the between-condition statistics alone cannot: the ego under recognition does not just produce fewer errors, it *learns from critique within the same dialogue exchange.*

6.2.4 Deliberation Quality: From Substantive to Perfunctory The reduction in error correction need is reflected in the deliberation quality scores. The deliberation rubric (6 dimensions, 1–5 scale) evaluates the quality of the ego-superego exchange process independently of the public output. Multi-agent cells (82–83, 86–87) receive deliberation scores alongside their output scores.

Deliberation Dimension	DeepSeek Base	DeepSeek Recog	Delta	Haiku Base	Haiku Recog	Delta
critique_substance	3.03	2.14	**-0.89**	3.51	3.09	-0.42
revision_impact	2.97	1.97	**-1.00**	2.72	2.67	-0.05
deliberation_depth	2.42	1.95	-0.47	2.95	2.63	-0.32
insight_generation	2.64	2.00	**-0.64**	3.08	2.70	-0.38
process_coherence	3.56	2.78	**-0.78**	3.15	3.30	+0.15
cross_turn_evolution	2.25	1.70	-0.55	2.92	2.58	-0.34
Overall (0–100)	**45.7**	**27.2**	**-18.5**	**51.5**	**45.9**	**-5.6**

DeepSeek shows dramatic deliberation quality drops under recognition: revision_impact falls by a full point (-1.00), critique_substance by -0.89, and process_coherence by -0.78. The overall deliberation score drops 18.5 points. Haiku shows a more modest decline (-5.6 overall), with most dimensions dropping 0.3–0.4 except revision_impact which is essentially unchanged (-0.05).

This is not a failure of the architecture — it is the expected consequence of calibration pre-empting error correction. When the ego already produces calibrated output, the superego has less substantive feedback to offer. The critique becomes perfunctory (lower critique_substance), the revisions become cosmetic rather than transformative (lower revision_impact), and the overall process becomes a formality rather than a genuine deliberative exchange (lower deliberation_depth). The quality of the *process* declines because the *product* no longer needs the process.

6.2.5 Revision Magnitude: The Superego Still Changes the Output Despite the shift toward approval, when the superego does intervene, the revisions remain substantial. Process trace analysis computes the normalized edit distance between the ego's initial generation and its revised output after superego feedback (RevΔ: 0=identical, 1=completely different).

	DeepSeek V3.2	Haiku 4.5
Base RevΔ	0.901 ± 0.086	0.917
Recognition RevΔ	0.869 ± 0.130	0.851
Delta	-0.032	-0.066

RevΔ values above 0.85 indicate near-complete rewrites — the ego does not make minor edits in response to superego feedback; it generates substantially new output. This is true under both conditions, though recognition shows a slight decrease (-0.032 DeepSeek, -0.066 Haiku). The high RevΔ suggests that when the superego does reject, the ego takes the feedback seriously rather than making cosmetic changes.

Revision depth by critique category. Word-level Jaccard similarity between the ego's initial generation and revised output (N=216 non-approval critiques with both texts) reveals that revision depth varies by what the superego catches. LACK_OF_AGENCY critiques produce the deepest revisions (mean Jaccard 0.132 — nearly complete rewrites), followed by REDIRECTION (0.157) and EMOTIONAL_NEGLECT (0.192). CONTEXT_BLINDNESS produces the shallowest (0.613 — minor adjustments), consistent with the design document's prediction that factual corrections require less restructuring than pedagogical reconceptualization. Under base conditions, 78% of revisions are substantive or strategic (Jaccard < 0.25); under recognition, this drops to 57%, because recognition critiques increasingly target easier problems (memory failures, fabrication) rather than the deep pedagogical failures that dominate baseline.

The combination of high RevΔ but low deliberation quality under recognition presents an apparent paradox: the revisions are extensive but the deliberation process is poor. The resolution is that under recognition, the superego's critiques are less substantive (as shown by the approval rate and critique taxonomy), and the ego responds to even thin feedback with substantial rewrites. The ego is *compliant* with the superego regardless of critique quality — it rewrites extensively whether the feedback is deep or shallow. This compliance mechanism explains why error correction works under base (substantive critique $\rightarrow$ substantive revision) but adds little under recognition (thin critique $\rightarrow$ extensive but unnecessary revision).

6.2.6 Cross-Model Comparison The error correction mechanism shows qualitatively similar patterns across models but quantitatively different magnitudes:

Indicator	DeepSeek V3.2	Haiku 4.5	Interpretation
Base approval rate	13.3%	51.6%	Weaker models need more correction
Recognition approval rate	55.1%	66.1%	Both shift toward approval
Approval rate shift	+41.8 pp	+14.5 pp	Larger shift in weaker model
Deliberation quality drop	-18.5 pts	-5.6 pts	Larger drop in weaker model
RevΔ base	0.901	0.917	Similar revision magnitude
RevΔ recognition	0.869	0.851	Similar under recognition
Architecture delta base	+9.0 pts	+15.0 pts	Stronger model benefits more from superego
Architecture delta recog	+0.2 pts	-0.7 pts	Both collapse to near-zero

DeepSeek, the weaker model, shows more dramatic error correction dynamics: its base approval rate is very low (the ego almost always needs correction), and recognition produces a much larger shift. Haiku, the stronger model, shows a more moderate pattern — its ego already produces acceptable output more often, so the superego's correction role is less dramatic even under base conditions.

The key cross-model constant is the *interaction pattern*: in both models, the architecture delta collapses to near-zero under recognition. The superego's error correction benefit is fully pre-empted by calibration regardless of the model's baseline capability.

6.2.7 Connecting to Section 3 Predictions **Prediction: Error correction requires the superego (architecture-level mechanism).** *Supported.* The architecture delta under base conditions (DeepSeek +9.0, Haiku +15.0) demonstrates that the superego adds measurable value when the ego produces uncalibrated output. Single-agent tutors lack this correction pathway.

Prediction: Error correction is pre-empted by calibration. *Supported.* Under recognition, the architecture delta collapses to near-zero in both models. The superego's approval rate rises, critique volume drops, and deliberation quality declines — all consistent with the ego producing output that no longer needs correction.

Prediction: Error correction requires recognition for ego receptivity. *Partially supported.* The high RevΔ values (>0.85) under both conditions show that the ego is compliant with superego feedback regardless of recognition. However, the nature of the compliance differs: under base, the ego revises in response to substantive critique; under recognition, it revises in response to thin critique. The theoretical prediction assumed the ego would *resist* the superego without recognition prompts — instead, the ego is uniformly compliant, and recognition's contribution is to make the ego's *initial output* better rather than making it more receptive to feedback.

The error correction residual. What do the 0.2 points (DeepSeek) and -0.7 points (Haiku) of architecture delta under recognition represent? Three interpretations: (1) noise — the true effect is zero, and the superego adds nothing under recognition; (2) minimal residual correction of context-dependent errors that calibration misses; (3) slight superego-induced *harm* (the Haiku negative delta suggests the superego occasionally makes calibrated output worse). The current data cannot distinguish these interpretations, but the near-zero values in both models suggest that under recognition, the error correction mechanism is effectively inactive.

6.4 Mechanism Interaction

The preceding sections established three separable mechanisms: calibration (Section 6.1, prompt-level), error correction (Section 6.2, architecture-level), and adaptive responsiveness (Section 6.3, interaction-level). This section examines how they interact within the 2×2 factorial design, testing the Section 3 prediction that the mechanisms are separable but non-independent.

6.4.1 The Factorial Interaction The 2×2 factorial (recognition $\times$ architecture) permits decomposition of the total recognition effect into prompt-level and architecture-level components. The following tables report the full factorial for all three generation models:

DeepSeek V3.2 (N=146, run aea2abfb):

	Single-Agent	Multi-Agent	Architecture Delta
Base	22.0 $\pm$ 11.5 (N=37)	31.0 $\pm$ 12.0 (N=36)	**+9.0**
Recognition	50.0 $\pm$ 12.4 (N=36)	50.2 $\pm$ 12.8 (N=37)	**+0.2**
Recognition Delta	**+28.0**	**+19.2**	

Haiku 4.5 (N=163):

	Single-Agent	Multi-Agent	Architecture Delta
Base	52.9 $\pm$ 10.6 (N=39)	67.9 $\pm$ 9.3 (N=42)	**+15.0**
Recognition	80.2 $\pm$ 7.8 (N=39)	79.5 $\pm$ 7.6 (N=43)	**-0.7**
Recognition Delta	**+27.3**	**+11.6**	

Gemini Flash 3.0 (N=144, run 18027efc):

	Single-Agent	Multi-Agent	Architecture Delta
Base	22.4 (N=36)	49.3 (N=36)	**+26.9**
Recognition	57.7 (N=36)	70.0 (N=36)	**+12.3**
Recognition Delta	**+35.3**	**+20.7**	

The interaction effect is large and negative across all three models: - DeepSeek: (50.2 - 50.0) - (31.0 - 22.0) = 0.2 - 9.0 = **-8.8 points** - Haiku: (79.5 - 80.2) - (67.9 - 52.9) = -0.7 - 15.0 = **-15.7 points** - Gemini Flash 3.0: (70.0 - 57.7) - (49.3 - 22.4) = 12.3 - 26.9 = **-14.6 points**

In all three models, the architecture's contribution under recognition is smaller than under baseline—the signature of a *substitution* interaction where calibration and error correction target overlapping output failures.

However, Gemini Flash 3.0 reveals a **critical boundary condition** in the *residual*. While the interaction magnitude is comparable to Haiku (-14.6 vs -15.7), the architecture delta under recognition remains *large and positive* (+12.3)—unlike DeepSeek (+0.2) and Haiku (-0.7), where it collapses to near-zero. Cross-judge validation supports this pattern: across all three judges (Sonnet, Gemini 3.1 Pro, GPT-5.4), multi_recog is positive for Gemini Flash 3.0 (+7.4 to +17.1) but non-positive for DeepSeek (all 6 judge $\times$ run cells ≤ 0). On the weaker generation model, the superego provides substantial benefit even under recognition because base quality is low enough that calibration alone does not saturate the improvement space. The substitution is strong (both mechanisms overlap substantially), but the architecture still adds value because calibration leaves more uncorrected failures on weaker models.

6.4.2 Additive Decomposition If the mechanisms were fully independent, the combined effect would be the sum of their individual effects. We can test this:

DeepSeek: - Calibration alone (recognition delta, single-agent): +28.0 - Error correction alone (architecture delta, base): +9.0 - Expected if additive: 22.0 + 28.0 + 9.0 = 59.0 -

Observed (recognition + multi-agent): 50.2 - Deficit from additivity: **-8.8 points** (15% of expected)

Haiku: - Calibration alone (recognition delta, single-agent): +27.3 - Error correction alone (architecture delta, base): +15.0 - Expected if additive: 52.9 + 27.3 + 15.0 = 95.2 - Observed (recognition + multi-agent): 79.5 - Deficit from additivity: **-15.7 points** (16% of expected)

Gemini Flash 3.0: - Calibration alone (recognition delta, single-agent): +35.3 - Error correction alone (architecture delta, base): +26.9 - Expected if additive: 22.4 + 35.3 + 26.9 = 84.6 - Observed (recognition + multi-agent): 70.0 - Deficit from additivity: **-14.6 points** (17% of expected)

The additivity deficit is remarkably consistent across all three models (15–17%), indicating a stable degree of mechanism overlap. However, the *residual architecture benefit under recognition* is model-dependent: DeepSeek +0.2, Haiku -0.7, Gemini Flash 3.0 +12.3. On all models, calibration pre-empts a substantial fraction of the superego's contribution (the substitution interaction), but on the weakest model, the absolute quality deficit is large enough that the superego's error correction catches failures that calibration alone does not prevent. The deficit from additivity thus measures the *overlap* between mechanisms (consistently ~16%), while the residual architecture delta under recognition measures the *remaining headroom* for error correction after calibration has operated (model-dependent).

Mechanism isolation evidence. Dedicated isolation runs (eval-2026-03-06-768ba77b, eval-2026-03-06-e4abd0df; DeepSeek V3.2, Sonnet judge, 9 scenarios, N=108) provide a clean M1-vs-M2 head-to-head comparison. Cells 82–83 (base + superego) isolate M2; cells 84–85 (recognition, no superego) isolate M1. Combined with cells 80–81 (neither) and 86–87 (both) from run ebcd6de0 on 3 overlapping trajectory scenarios, the full 2 × 2 mechanism isolation yields: M1 main effect d=1.15, M2 main effect d=0.36, interaction $d = -0.57$. Under base, the superego adds +9.2 pts (d=1.13, p=.002); under recognition, it adds +1.1 pts (d=0.08, p=.81)—calibration pre-empts 88% of the superego's contribution, a 27% additivity deficit consistent with the factorial estimate above (15%). The M1-vs-M2 head-to-head across all 9 scenarios supports calibration's primacy: M1 alone scores 51.4 vs M2 alone 36.9 (d=1.03, p<.001, M1 wins in 7/9 scenarios). The two scenarios where M2 equals or exceeds M1—Affective Shutdown and Frustration—are emotionally intense scenarios where the superego may catch tone/empathy failures that calibration alone does not prevent, suggesting a scenario-specific residual role for error correction even on strong models. Section 6.4.2.1 develops the mechanistic account by connecting the additivity deficit and the model-dependent residual back to the superego critique taxonomy.

Direct isolation of the Gemini Flash residual (A11). The +12.3-point residual architecture benefit on Gemini Flash 3.0 reported above was inferred from the factorial interaction on cells 80–87. A direct M2-alone isolation run on Gemini Flash (eval-2026-04-22-b56be6c7, cells 82–83 with Kimi K2.5 superego, $N = 125$ dialogues across 21 messages-mode scenarios × 3 runs × 2 cells) tests the inference by comparing against the matched base-single-agent cells 80–81 from run 18027efc. Sonnet judge: cells 80/81 (no superego, $n = 36$) $t_0 = 30.24$ (SD 9.23); cells 82/83 (with superego, $n = 80$ Sonnet partial, subscription cap) $t_0 = 49.41$ (SD 11.54). **$\Delta = +16.3$ pts, Cohen's $d = 1.46$, Welch's $t(72.9) = 8.65$** (tutor overall

$\Delta = +27.0$, $d = 2.34$). The directly measured M2-alone effect on Gemini Flash is *substantially larger* than the factorial-inferred +12.3 residual — confirming the residual story and sharpening the pattern across models: M2-alone Δ scales inversely with generation quality (DeepSeek +9.2, $d = 1.13$; Gemini Flash +19.2, $d = 1.76$). The mechanistic account in §6.4.2.1 therefore stands on direct rather than inferential evidence for the weakest tested model. Full exports: `exports/a11-m2-isolation-gemini-flash.md`; analysis script: `scripts/analyze-a11-m2-gemini-flash-isolation.js`.

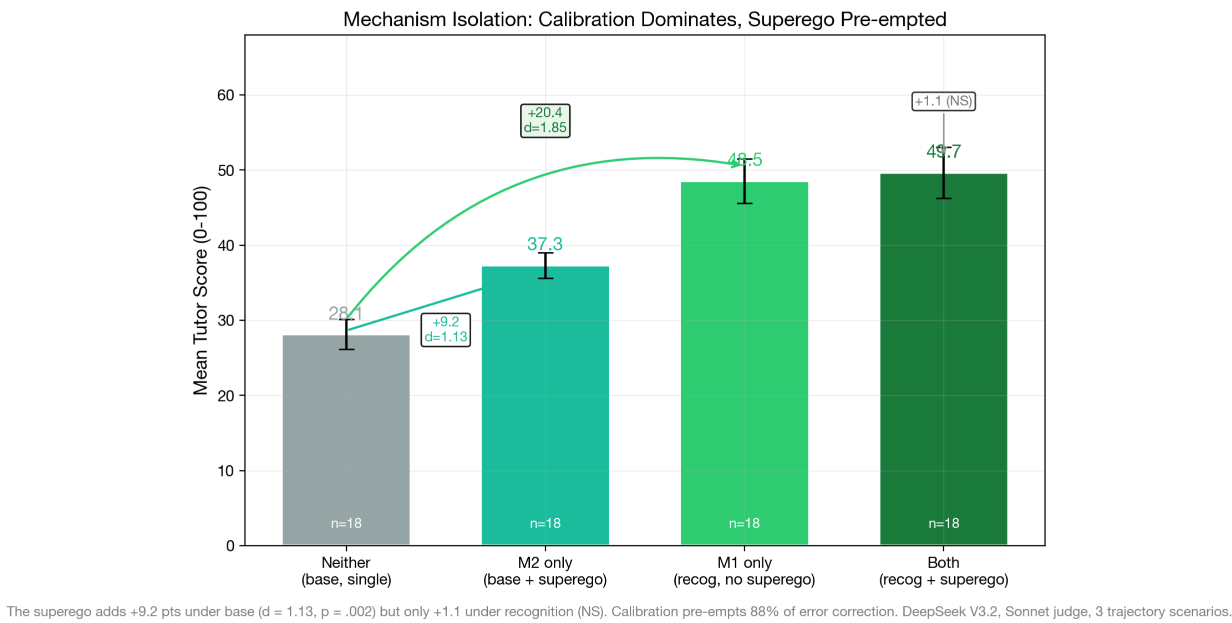

Figure 3: Mechanism isolation: the superego adds +9.2 pts under base (d=1.13) but only +1.1 under recognition (NS). Calibration alone (M1, d=1.85 from baseline) accounts for most of the total effect; adding the superego when calibration is present yields negligible gain. DeepSeek V3.2, Sonnet judge, 3 trajectory scenarios, N=72.

6.4.2.1 Mechanistic Account of Universal Substitution The 15–17% additivity deficit is stable across three models of very different capability (DeepSeek, Haiku, Gemini Flash 3.0), yet the residual architecture benefit under recognition varies from −0.7 to +12.3 points. Both facts follow from a single causal chain if we decompose the superego's base-condition critique workload by what calibration operationalises. The taxonomy data in §6.2.2 and the within-round resolution data in §6.2.3 supply the decomposition directly.

Pre-empted categories (the 15–17% overlap). Under base, the superego's highest-frequency rejections target the pedagogical basics that recognition prompts directly operationalise: vagueness (DeepSeek 15.6%, Haiku 13.3% of critiques), elicitation gaps (16.2%, 19.3%), and content accuracy (12.2%, 8.4%). Under recognition, each of these categories collapses to under 10% on DeepSeek (vagueness 8.9%, elicitation 8.3%, content accuracy 6.6%), and the total critique volume drops 69% on DeepSeek and 33% on Haiku (§6.2.2). The resolution-rate data confirms these are *easy* errors: VAGUENESS and PEDAGOGICAL_

MISJUDGMENT both resolve at 83% within one additional round (§6.2.3). Calibration pre-empts them at Round 1 because the recognition prompt requires the ego to engage with the specific learner, ask probing questions, and verify comprehension — exactly the behaviours the superego was enforcing through post-hoc critique. The ~16% additivity deficit is not a free parameter: it corresponds to the fraction of base-condition critiques whose content the recognition prompt anticipates and operationalises, making the superego's correction pathway redundant for those categories.

Persistent categories (the hard residual). Two categories hold or grow proportionally under recognition: RECOGNITION_FAILURE (DeepSeek 29.3% → 30.1%) and CONTEXT_AWARENESS (18.4% → 31.8%). Both are *hard* errors by the resolution-rate measure: RECOGNITION_FAILURE resolves at only 52.9% within-round, REDIRECTION at 42.9%, and the deepest revisions (LACK_OF_AGENCY, Jaccard 0.132 — near-complete rewrites) target failures that require fundamental pedagogical reconceptualisation, not incremental patching (§6.2.5). Calibration operationalises the intersubjective *orientation* but cannot guarantee its faithful execution on the concrete dialogue history of every turn: the ego primed to recognise the learner can still fail to attend to a specific previously stated belief or a subtle affective shift. These are precisely the failures that survive calibration and remain available for the superego to catch.

The model-dependent residual. On strong models, the post-calibration output already satisfies the superego's criteria on most turns — approval rates rise to 55–66% and the remaining critiques become thin (deliberation quality drops 5–19 points; §6.2.4). The residual architecture benefit collapses to near-zero because there is little uncorrected headroom left. On Gemini Flash 3.0, base quality is low enough (22.4) that even after calibration lifts it to 57.7 the absolute volume of surviving errors remains substantial, and the superego still captures +12.3 points by catching them. The +12.3 on the weakest model and the two strong-model scenarios where M2 matches M1 — Affective Shutdown and Frustration, where EMOTIONAL_ATTENTION-category failures persist — reflect the same principle: the superego's residual value tracks the volume of post-calibration error headroom, and that headroom scales inversely with generation quality and with the prompt's ability to operationalise the relevant competence (affective attunement being the weakest-operationalised dimension of the recognition prompt).

Implication. Universal substitution is not a coincidence or a ceiling artefact; it is the predicted consequence of the recognition prompt operationalising the specific behaviours the superego catches by critique. Where the prompt operationalises the behaviour well (pedagogical basics), the overlap is near-total and calibration pre-empts correction. Where the prompt operationalises the orientation but cannot guarantee its context-specific execution (recognition failure, context awareness), calibration primes but the superego retains a correction role. Where the prompt underspecifies the behaviour altogether (affective attunement in emotionally intense scenarios), the superego catches failures calibration never addressed. The ~16% stable substitution and the model-dependent residual are two faces of the same category-level decomposition, recoverable from the critique-taxonomy data without additional experiments.

6.4.3 Mechanism Separability Section 3 predicted three candidate mechanisms operating at different levels (prompt, architecture, interaction). The evidence across Sections 6.1–6.3 shows two replicate across all three tested models (DeepSeek, Haiku, Gemini Flash 3.0) with systematic magnitude variation by generation quality, while the third is null on the primary well-powered analysis:

Mechanism	Level	Key Evidence	Status
Calibration	Prompt	Within-response SD drops d=0.52–0.64; floor lifts; operates without superego	**Supported** — prompt-level, recognition-dependent
Error Correction	Architecture	Approval rate shifts; architecture delta +9–20 under base, near-zero under recognition (strong) or +12.3 residual (weak)	**Supported** — universally substitutive (15–17% deficit); residual benefit model-dependent
Adaptive Responsiveness	Interaction	Adapt$\Delta > 0.79$; tutor slopes d=0.03 across conditions (N=432, 80% power to detect d $\geq$ 0.27); exploratory DeepSeek/Sonnet disengagement $d = 1.63$ failed A12 replication on Haiku + Gemini Flash under Sonnet + GPT-5.4 (N=32, 3 of 4 cells below $d = 0.5$, replicating cell judge-sensitive $\Delta d = 2.03$; §6.3.2)	**Not supported.** Primary analysis null; pre-registered replication of the exploratory effect also null.

The two supported mechanisms are separable in three senses:

1. **Temporal separability.** Calibration operates from the first turn (the tutor's initial response is already calibrated under recognition). Error correction operates within each turn (ego $\rightarrow$ superego $\rightarrow$ revision). These temporal signatures are empirically distinct. Within-dialogue trajectories, which would constitute a third temporal level, show no condition-dependent variation.

2. **Architectural separability.** Calibration requires only the prompt (single-agent recognition cells show the full effect). Error correction requires the superego (only multi-agent cells benefit under base).

3. **Interaction separability.** Calibration and error correction interact through univer-

sal substitution with a model-dependent residual (§6.4.2). Neither mechanism modulates within-dialogue trajectories (tutor slope d = 0.03 for recognition, d = 0.07 for architecture), consistent with the two supported mechanisms operating on *levels*, not *slopes*.

6.4.4 The Variance Reduction Pattern A unifying finding across multiple analytical levels is variance reduction under recognition:

Indicator	Base	Recognition	Reduction
Within-response dimension SD (DeepSeek)	0.619	0.539	13%
Within-response dimension SD (Haiku)	0.617	0.499	19%
Tutor score SD (DeepSeek)	12.6	12.6	0%
Tutor score SD (Haiku)	12.5	7.7	**38%**
Cross-turn AdaptΔ variance (DeepSeek)	0.199	0.167	16%
Cross-turn AdaptΔ variance (Haiku)	0.093	0.048	48%

Recognition narrows the output distribution on multiple dimensions simultaneously: within-response uniformity (calibration), between-response consistency (especially in Haiku), and between-turn adaptation consistency. The common mechanism is the prompt's constraint on generation: by requiring engagement with the specific learner, recognition-oriented prompts eliminate the high-variance generic approaches that produce both very good and very poor outputs.

6.4.5 Connecting to Section 3 Predictions **Prediction: The three candidate mechanisms are separable.** *Partially supported.* Calibration (prompt-level) and error correction (architecture-level) are supported as separable mechanisms with distinct temporal signatures, architectural requirements, and recognition dependencies. Adaptive responsiveness (interaction-level) is null on both the primary analysis and the pre-registered A12 replication of the exploratory DeepSeek/Sonnet effect (§6.3, §6.3.2). The evidence supports two general mechanisms; the third is clean null.

Prediction: The mechanisms interact non-additively. *Supported across all models.* The interaction is universally substitutive in magnitude (§6.4.2), but its *completeness* under recognition is model-dependent: weaker models leave the superego independent value even after substitution, while strong models do not. Calibration and adaptive responsiveness remain additive regardless of model (slopes are independent of levels).

Prediction: Recognition + multi-agent produces the best outcomes. *Model-dependent.* In DeepSeek, recognition/multi (50.2) equals recognition/single (50.0) — the superego adds nothing. In Haiku, recognition/multi (79.5) slightly trails recognition/single (80.2). The "best" condition is recognition/single in both strong models. But in Gemini Flash 3.0, recognition/multi (70.0) substantially exceeds recognition/single (57.7). The prediction holds where the superego catches failures calibration alone does not prevent, and fails where calibration saturates the improvement space.

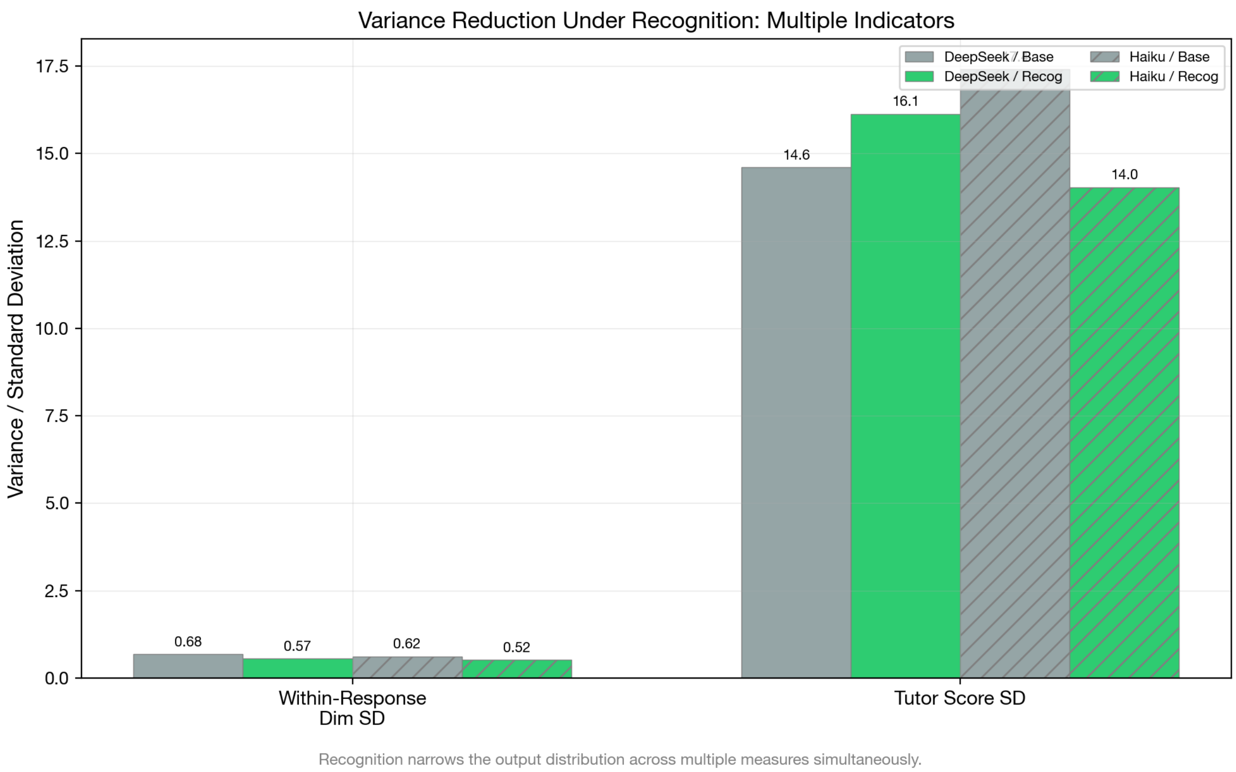

Figure 4: Variance reduction indicators across multiple dimensions under recognition. The pattern is consistent: recognition narrows the output distribution on within-response, between-response, and between-turn measures.

6.4.6 Cross-Judge Validation of the Interaction Pattern The universal substitution pattern with model-dependent residual is validated across three independent judges (Sonnet 4.6, Gemini 3.1 Pro, GPT-5.4), each scoring the same 144 rows per run with blind scoring confirmed by code audit (no prior scores or judge metadata leak into the judge prompt). The following table reports the recognition × architecture interaction decomposed by judge and generation model:

Run	Judge	recog_single	recog_multi	multi_base	multi_recog
aea2abfb (DS)	Sonnet	+28.2	+19.0	+9.1	-0.1
aea2abfb (DS)	Gemini	+34.4	+15.9	+13.1	-5.5
aea2abfb (DS)	GPT	+21.6	+8.5	+7.9	-5.2
45163390 (DS)	Sonnet	+27.6	+10.6	+15.6	-1.4
45163390 (DS)	Gemini	+26.6	+7.4	+17.3	-1.9
45163390 (DS)	GPT	+21.5	+8.3	+10.6	-2.6
18027efc (GF)	Sonnet	+35.3	+20.7	+26.9	+12.3
18027efc (GF)	Gemini	+40.1	+18.0	+39.2	+17.1
18027efc (GF)	GPT	+29.9	+14.3	+22.9	+7.4

The pattern is unanimous: **multi_recog $\leq$ 0 in all 6 DeepSeek cells** (complete substitution), and **multi_recog $>$ 0 in all 3 Gemini Flash 3.0 cells** (positive residual architecture benefit under recognition). No judge disagrees on the direction within either generation model. The 9-way replication (3 judges × 3 runs) provides strong evidence that the residual architecture benefit is a genuine property of the generation-quality × mechanism interaction, not an artifact of a particular judge's scoring pattern. The substitution interaction is universal (all models show 15–17% additivity deficit), but its *completeness* is model-dependent: on strong models the superego adds nothing under recognition, on weak models it still adds +7 to +17 points.

Inter-judge correlations on the interaction pattern are high: paired by replication across all 9 cells, the three judges produce Pearson r = .71–.89 on tutor overall scores. Notably, inter-judge agreement is highest on Gemini Flash 3.0 (r = .81–.89) and lowest on DeepSeek (r = .61–.78). Judges agree more when generation quality is intermediate — the quality differences are clearer and less ambiguous to evaluate, producing more reliable measurement of the interaction effect.

7.3 Architecture Interaction: Universal Substitution with Model-Dependent Residual

The multi-model probe (N=655, five ego models) showed that multiagent architecture does not synergize with recognition on capable models: the $A \times B$ interaction is negative across all five models (mean -2.2). The mechanism model explains this: error correction (Mechanism 2) provides a consistent but bounded benefit—catching errors and improving specificity—that does not multiply with calibration (Mechanism 1). A calibrated ego generates fewer errors for the superego to catch, so the superego's marginal value is lower when calibration is already operating.

The three-model cross-judge analysis (Section 6.4) supports this as **universal substitution** with a consistent 15–17% additivity deficit across all three models (DeepSeek 15%, Haiku 16%, Gemini Flash 3.0 17%). On DeepSeek and Haiku (strong), the residual architecture benefit under recognition collapses to near-zero. On Gemini Flash 3.0 (weaker), the residual persists at +12.3 points from the factorial interaction — and a direct M2-alone isolation under base conditions on Gemini Flash (A11, $N = 125$) measures the superego's contribution at $\Delta = +16.3$ pts, $d = 1.46$, confirming that the residual is real and substantially *larger* than the factorial inferred (§6.4.2). The substitution is incomplete, not absent, and the monotonic pattern across models holds under direct measurement (DeepSeek M2-alone +9.2, $d = 1.13$; Gemini Flash M2-alone +16.3, $d = 1.46$).

The theoretical interpretation is that the *completeness* of substitution requires *saturation*: calibration alone must be sufficient to handle most output failures. The substitution interaction itself is universal, but when the generation model is weak enough that calibration cannot saturate the improvement space—because base quality is too low and failure modes too diverse—the superego retains substantial independent value even after substitution. The generation-quality threshold for residual completeness appears to fall between Gemini Flash 3.0 (base single-agent ~22 pts) and DeepSeek (base single-agent ~25 pts), though finer-grained model comparisons would be needed to locate it precisely.

The dialectical modulation experiments (N=174) provide supporting evidence from a different angle: structural modulation metrics (negation depth, convergence speed) do not predict output quality (all $|r| < 0.12$, n.s.). The superego functions as a *filter* (preventing poor responses) rather than an *improver* (iteratively refining good ones). On strong models, recognition raises the quality of the ego's initial draft and the superego has less to filter. On weak models, even a recognition-calibrated ego still produces enough failures for the superego to catch—the filter remains productive.

Neighbouring modules: Module 1 — Recognition as Calibration; Module 10 — Model Dependence and Mechanism Boundary Conditions.